



# Understanding the influence of IT/OT Convergence on the adoption of Internet of Things (IoT) in manufacturing organizations: An empirical investigation

Ike C. Ehie<sup>a,\*</sup>, Michael A. Chilton<sup>b</sup>

<sup>a</sup> Operations and Supply Chain Management, College of Business, Kansas State University, Manhattan, Kansas 66502, United States

<sup>b</sup> Management Information System, College of Business, Kansas State University, Manhattan, Kansas 66520, United States

## ARTICLE INFO

### Article history:

Received 12 September 2019

Received in revised form

10 November 2019

Accepted 14 November 2019

Available online 22 November 2019

## 1. Introduction

The Internet of Things (IoT) is considered to be a network of smart devices that are connected to sense, monitor and interact both within a company and between the company and its supply chain. Its purpose is to enable agility, visibility, tracking and information sharing to facilitate timely planning, control and coordination of the supply chain processes (Ben-Daya et al., 2017). It is among the latest digital technologies that represents a paradigm shift in several business operations. In manufacturing, IoT is able to accomplish its purpose and provide solutions to numerous and varied challenges (Ellis et al., 2015). Two-thirds of manufacturing executives surveyed believe that IoT will increase their competitive advantage over time; however, most companies have limited understanding of IoT adoption or even how it might be applied in their business processes (Brandt, 2015). Thus, research is needed not just to identify determinants to IoT adoption in the manufacturing industry, but to help provide a roadmap for adoption as well.

IoT is comprised of four layers: (1) data collection, (2) transmission, (3) service and (4) an interface (Ben-Daya et al., 2017; Hofmann and Rüscher, 2017; Gubbi et al., 2013; Verdouw et al., 2013a; Borgia, 2014; Xu et al., 2014). The data collection layer is usually implemented with RFID antennas and tags; transmission is achieved using either wired or wireless means or a combination of both and can be either fixed or mobile; services provided include monitor and control functions; and interfaces either enable con-

nections with other devices and systems, or provide monitoring and control functions directly to users. As a key enabler of Industry 4.0 (Gurria, 2017), IoT networks can enhance manufacturing performance, especially in globalized and distributed environments (Bi et al., 2014). The emergence of IoT systems can effectively support next-generation manufacturing enterprises that extend from the beginning of the production line to and including warehouse operations. The ability to provide accurate data in real-time regarding the flow of materials and products is transforming the supply chain (Lee and Lee, 2015). Firms invest in IoT to redesign factory workflows, improve tracking of materials, and optimize distribution costs (Rodrigue et al., 2017). For example, both John Deere and UPS are already using IoT-enabled fleet tracking technologies to cut costs and improve supply efficiency (Gold, 2017; Horwitz, 2018). The true value of IoT for enterprises can only be fully realized when connected devices are able to communicate with each other and utilize the data produced to improve such systems as inventory tracking and control, customer support, and business intelligence and analytics. Gartner (2014) forecast that IoT will reach 25 billion units by 2020, up from 0.9 billion in 2009, and will impact the information available to supply chain partners and how the supply chain operates. IoT allows the use of intelligent networks of physical devices that may include sensors and actuators to communicate with one another directly or across both private and public networks to enable effective business decisions (Lee and Lee, 2015). Porter and Heppelmann (2015) argue that these smart-connected devices are changing the core functions of manufacturing—from new product development to manufacturing and/or from logistics to after-sales service.

By combining previous work in technology adoption with the unique capabilities of IoT and eliciting responses from professionals

\* Corresponding author.

E-mail addresses: [iehie@ksu.edu](mailto:iehie@ksu.edu) (I.C. Ehie), [mchilton@ksu.edu](mailto:mchilton@ksu.edu) (M.A. Chilton).

currently involved in manufacturing, we develop and test a framework that should provide a roadmap for effective IoT adoption. The framework centers on IT/OT convergence, which we define as the integration of information technology (IT) systems used for data-centric computing with operational technology (OT) systems designed for monitor and control purposes. We developed our framework by integrating results from previous studies on technology, organization and environment (TOE) (Tornatzky et al., 1990), diffusion of innovation (DOI) (Rogers, 2003), and organizational readiness (Iacovou et al., 1995). We found that certain factors rooted in technology (IT governance, IT/OT infrastructure and interoperability, and to a lesser extent staff collaboration) significantly influenced IT/OT convergence, which in turn, can improve the chances for successful IoT adoption. Against this backdrop, this study takes on a socio-technical perspective to develop a framework that would assist manufacturing companies in evaluating and adopting IoT systems in their operations.

This paper is organized as follows. In the next section we review the historical background of IoT by providing some definitions and identifying enabling technologies based on an examination of the extant literature. We then present our research framework in the next section and discuss its logic and offer six hypotheses generated by our model. The results are presented in the next section and an analysis of them. We finish with our conclusions, limitations of the study and we suggest some areas for further research.

## 2. Literature Review

Manufacturing is reported to have significant importance to the prosperity of nations with over 70% of the income variations in 128 countries explained by the differences in manufactured product export data alone (World Economic Forum (WEF) Deloitte Touche Tohmatsu (Firm), 2012). Manufacturing is currently evolving into what is known as “Industry 4.0,” in which IoT is causing a major paradigm shift that will have profound implications on the entire supply chain (Ben-Daya et al., 2017). Smart manufacturing thus enables better decisions that can result in more efficient operations not just within the factory, but throughout the supply chain, because it is based on real-time information (Buyukozkan and Gocer, 2018). The benefits of IoT include such things as enhanced factory visibility, better connections within the supply chain, proactive maintenance, improved quality that extends beyond the factory, greater sustainability, and the ability to conduct planning and scheduling based on data collected in real time (Ben-Daya et al., 2017). This promise of IoT technologies has gained wide attention and has resulted in a large number of applications in many fields (Schlick et al., 2013). Verdouw et al. (2013b) argue that IoT enables the virtualization of supply chains in which the buyer is able to track and trace goods as they move through the supply chain as well as perform advanced quality control and planning. Ng et al. (2015) proposed a model to integrate data collected from IoT into strategic planning for product assortments. Yu et al. (2015) studied the impact of IoT on suppliers and recommend the adoption of IoT technologies that would offer more flexibility. Decker et al. (2008) have identified several benefits of IoT regarding sourcing. While IoT promises to provide valuable real-time visibility to the supplier (Liu et al., 2016), it comes at a cost. Decker et al. (2008) developed a simple linear cost model to analyze the impact of the cost of sensors and alerts on the unit purchase cost. IoT applications constitute the interface between the user and the devices, allowing the presentation of data in an intuitive way, while at the same time identifying problems and suggesting solutions (Lee and Lee, 2015).

Tan and Wang (2010) provided a description of IoT and addressed some of its essential issues, including its architecture

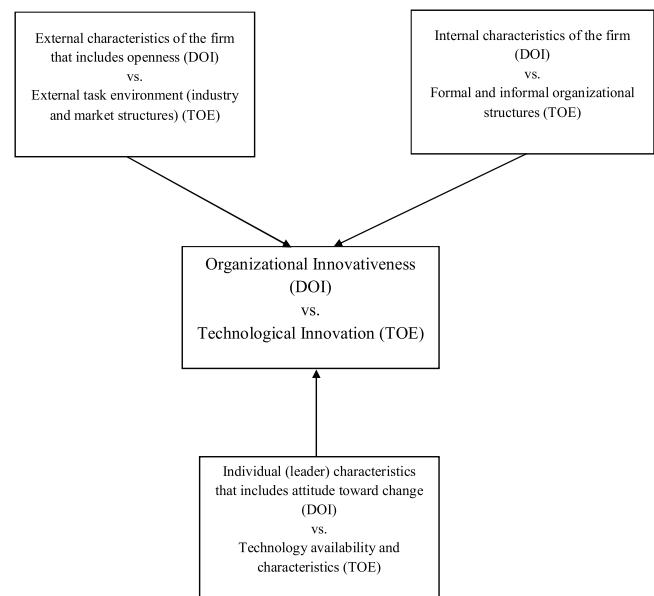


Fig. 1. Similarities between the TOE and the DOI frameworks.

and interoperability among disparate systems. Haller et al. (2009) identified major technical issues, including internet scalability, heterogeneity, identification, and addressing. Additionally, investigation into the user acceptance of information technology (IT) has always been an important issue in information systems technology (Bandyopadhyay and Sen, 2011). Other studies (Liu et al., 2014; Yao et al., 2017) provide comprehensive reviews and applications of IoT in manufacturing and supply chain management.

The basis for our study is primarily taken from two existing theories: the technology, organization, and environment (TOE) and diffusion of innovation (DOI) models. The TOE model developed by Tornatzky et al. (1990) helps to explain how an organization will implement technology innovation. The technological context describes the internal and external technologies of the firm; the organizational context describes the formal and informal structures and measures by which an organization describes itself; and the environmental context describes the setting in which the firm conducts business, including its communication channels. TOE has proven to be a parsimonious model with high explanatory power of the variance in adoption of new technologies across a wide variety of contexts (Ha and Stoel, 2009). The DOI model developed by Rogers (2003) describes organizational innovativeness as comprised of individual (leader) characteristics, internal organizational structures and the external features of the environment. Fig. 1 depicts the similarities between the two frameworks. The TOE and the DOI models as originally presented and later adapted in IT adoption studies (Tornatzky et al., 1990; Rogers (2003)) provide useful analytical frameworks that can be used in studying the adoption of different types of IT innovations. Both models emphasize individual characteristics as well as internal and external characteristics of the firm as being drivers for organizational innovativeness which can lead to effective IT adoptions. Although both are similar in technology and organization contexts, TOE is focused more on explaining the environmental context that presents both constraints and opportunities for technological innovations. Both TOE and DOI identified firm size to influence organizational innovativeness (Hsu et al., 2006). Additional studies have combined the two models to help understand and explain IT adoption (Iacovou et al., 1995; World Economic Forum (WEF) Deloitte Touche Tohmatsu (Firm), 2012). These models have generated an extensive number of

studies that explore both organizational and environmental factors in technology acceptance and adoption.

With the advent of digital communications and networks, more inter-organizational systems (IOSs) are being designed and used to integrate the operations among multiple businesses. This is especially true with IoT networks. However, such integration increases the complexity of innovations (Hsu et al., 2006). The two frameworks discussed so far focus on the individual firm and may not capture the effects that inter-organizational systems (IOS) have on the adoption of IT innovations. To correctly understand IoT adoption, we need to add the effects of the supply chain into the model. To do this, we augment the TOE and DOI models with the Iacovou et al. 1995 framework to add the inter-organizational systems (IOSs) dimension of IoT adoption to our model.

Iacovou et al. (1995) analyzed seven case studies to examine the impact of adoption and implementation of electronic data interchange (EDI) systems on small firms. Their framework identified external pressure, organizational readiness and perceived benefits as important factors affecting IT adoption. Since EDI uses communications networks among organizations in which pressure from trading partners plays a critical role in its adoption and use, we added this to our model to provide the link to the supply chain that TOE and DOI do not include. Several industry surveys (Brandt, 2015; Gartner, 2014) have shown that benefits can be derived from IoT adoption in support of the Iacovou et al. (1995) framework.

Our framework draws from both TOE and DOI, and includes the model developed by Iacovou et al. (1995) to include inter-organizational pressures, which is commonplace among members in a supply chain. The TOE framework helps to explain how technological change comes about within an organization. It is dependent upon the availability and characteristics of the technology, the communication structures that link disparate parts of an organization and the nature of the market in which the organization competes (Kuan and Chau, 2001). Diffusion of Innovations theory is based primarily on the ways in which the attributes of an innovation are communicated and perceived through a social system over time (Rogers, 2003), and therefore adds a social aspect to what is otherwise a technical model. Our model begins with these foundations but looks deeper into the specific factors that we theorize can better explain and later enable the adoption of IoT technology. These factors are based upon the communications channels uncovered by DOI theory and which are found to enhance technological change according to TOE theory. We suggest that these factors are key enablers in the adoption of IoT systems. Given this background, our model attempts to show that they can influence IoT adoption by bringing together IT and OT systems, staff and technology. We refer to this phenomenon as IT/OT convergence, and we predict that this variable will reflect the enablers, and doing so can provide additional explanation of the variance in IoT adoption. Our initial model shows IT/OT convergence as a prerequisite to IoT adoption; however, we suspect that it may play an additional role in the model as a mediator. We therefore investigate this further by testing for mediation between the enablers and the ultimate dependent variable, IoT adoption. We suggest that knowledge of this mediation can provide helpful assistance to practitioners as they engage in adoption of IoT systems.

### 3. Hypotheses Development

The framework proposed in this study utilizes four factors shown to be enablers of adoption of innovative technology taken from the literature. The construct of interest, IoT adoption, is our dependent variable; however, because IoT systems represent a form of operational technology (OT) that relies on the underlying information technology, we include measures of IT/OT conver-

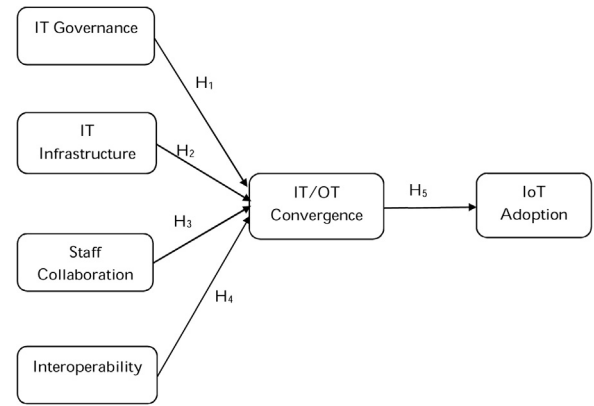


Fig. 2. Proposed research framework.

gence, because it has been identified as a key factor in the adoption of smart systems (Porter and Heppelmann, 2015). While there isn't much in the academic literature to describe this convergence and its effects, anecdotal evidence gathered from practitioners indicates that this is occurring. As IT is used more and more for data exploration and analysis, and as OT provides the signals and data to be used and analyzed, it appears that the two are beginning to converge and blur the lines of distinction between them. It follows then that this convergence may affect IoT adoption (Starbuck, 1976; Oliveira and Martins, 2011). Our final model places IT/OT convergence between the original four factors and the dependent variable, because we suspect not simply a direct effect on IoT Adoption but a mediating one as well. The final model is shown in Fig. 2.

#### 3.1. IT Governance

The phrase "IT Governance" represents the framework used to establish decision rights and accountabilities that influence desired outcomes in the use of IT (Li, 2006). IT governance draws upon corporate governance principles to manage the use of IT to achieve corporate performance goals (Chau and Tam, 1997). IT governance can influence decisions on the location, distribution and pattern of managerial responsibilities and control that ultimately affect how IT resources are applied and implemented (Haider, 2012). Effective IT governance ensures that there are controls in place to ensure that IT decisions and subsequent investments are carried out positively, equitably and are devoid of personal biases. Its goal is to ensure that the deployment of new technologies are aligned with organizational goals and business strategy. The governance issue is a necessary component of IT/OT convergence because, as Haider (2012) argues, operational technology is used for control and monitoring while utilizing the IT infrastructure, but the two technologies are considered separate and distinct. As these technologies begin to converge, a single governance policy should be in place to effectively manage both. Scales included in the IT governance include top management role in enabling IT/OT convergence (Ye and Wang, 2013), strategic redefinition of organizational units that lends to IoT adoption (DePietro et al., 1990), and patterns of managerial responsibilities and control in the organization (Loh and Venkatraman, 1992; De Haes and Van Grembergen, 2005). Hence, we offer the following hypothesis:

**H1.** Effective IT governance structure is positively related to the IT/OT convergence.

#### 3.2. Interoperability

We define interoperability as the ability of disparate systems to communicate. It has been identified as the biggest technical chal-

lenge to IoT adoption (Gubbi et al., 2013; Porter and Heppelmann, 2015), because it not only affects new IoT systems, but it must also integrate different types of data for analysis. An IoT system must interact with objects in our physical world (Porter and Heppelmann, 2015) and be able to transmit and receive such data and act upon it regardless of where the data originates. Further, Aloï et al. (2018) claim that all common objects in use today will be interconnected with monitor and control capabilities. This creates significant challenges in interoperability among devices. They introduce a smartphone-based mobile gateway to address these issues. To facilitate cross communication among devices, the use of IoT “hubs” has been advocated. Other solutions have also been suggested. For example, Luo et al. (2018) discuss network integration in smart factory environments and propose integrating two main networks used for data transfer to address the issues of data transmission capability and power consumption. An IoT hub could facilitate increased interoperability by using standardized software interfaces and protocols (e.g., using XML files as interfaces). The use of large scale IoT hubs has the potential to create an IoT ecosystem of global proportions. Interoperability on a global scale is a major challenge for two reasons: the heterogeneity of stakeholders whose systems vary widely, and the development of non-standard and often proprietary communications protocols. Developing complex communications protocols that are standardized and allow for seamless installation of new IoT systems that employ plug-and-play modules can lead to significant business opportunities (Khandwalla, 1970). Inter-operability refers to the ability of systems within an organization to be able to link to other systems, both within the organization and with other organizations, especially those in the supply chain. It is influenced by the interconnectedness of the organization, which is predicated on the degree to which units in the social strata of an organization are linked by interpersonal and technological networks (Oliveira and Martins, 2011; Li, 2006). Items designed to capture interoperability include compatible and seamlessly connected IT systems with suppliers (Buyukozkan and Gocer, 2018), extent of technical integration across disparate systems (Ha and Stael, 2009) and leveraging IT capabilities among systems across the supply chain (Li, 2006). We therefore proffer the following hypothesis:

**H2.** Improved systems interoperability of IoT protocols has a positive influence on IT/OT convergence.

### 3.3. Staff collaboration

IT/OT staff collaboration refers to the level of knowledge and expertise that the staff members possess and the degree to which an organization follows rules and procedures (Starbuck, 1976). The use of information technology to facilitate staff collaboration has become an important theme in information systems research and practice (de Villiers, 2015). Emerging technologies, such as groupware products and the Internet, have improved the way in which people communicate and collaborate in the organization. Broader social and organizational context within which information technology is implemented has a profound influence on the way in which IT/OT is used and communicated (Boynton et al., 1992). Tan and Wang (2010) found that successful collaboration among supply chain echelons positively affect a firm’s market performance. This type of collaboration is essential to developing the talent base in manufacturing needed for the adoption of IoT. Items designed to measure competence and collaboration include recruitment of appropriate talent needs of the organization (Khandwalla, 1970), resolving technical issues (de Villiers, 2015), upgrading legacy systems [reference] and the ability to link operations data with business analysis functions (Boynton et al., 1992). We suspect that

IT and OT staff collaboration is integral to effective IT/OT convergence and offer the following hypothesis:

**H3.** IT/OT Staff collaboration has a positive influence on IT/OT convergence

### 3.4. IT Infrastructure

IT infrastructure is the enabling foundation capable of delivering IoT services to users where and when needed Duncan (1995) describes IT infrastructure as a set of shared, tangible IT resources forming a foundation for business applications. These include platform technology (hardware and operating systems), network and telecommunication technologies, data, and core software applications. Weill (1993) notes that IT infrastructure is the foundation for capabilities across business and functional units. Davenport and Linder (1994) refer to IT infrastructure as that part of the organization’s information capacity that is intended to be shared and the foundation upon which the specific business activities and computer applications are built. Rockart et al. (1996) refer to IT infrastructure as the telecommunication, computers, software, and data that is integrated and interconnected so that all types of information can be expeditiously routed through the network and redesigned processes. Broadbent and Weill (1997) and Henderson and Venkatraman (1994) separate the IT infrastructure into IT architecture (applications, data, and technology configuration) and IT skills (knowledge and capabilities required to manage the IT resources within the organization). IT infrastructure is not a “nice-to-have” thing but rather a key driver for the entire business that requires proper analysis, design and execution plan for aiding and enabling the digital technology platform [52 To the extent that an organization is able to adapt their current infrastructure, organizations may find IoT systems easier to implement. Organizations found to change their infrastructure easily will find IT/OT convergence happen more quickly. To assess the effects of IT infrastructure on adoption of IoT systems, we adopted the following scales: the extent to which systems are decentralized across the organization (Broadbent and Weill (1997)), degree to which new technologies are adapted to older technologies (Henderson and Venkatraman, 1994), and the ability of the company to cope with frequent changes in production technology (Henderson and Venkatraman, 1994).

We therefore offer the following hypothesis regarding manufacturers:

**H4.** Flexibility in IT infrastructure has a positive effect on IT/OT convergence.

### 3.5. IT/OT convergence

IT/OT convergence refers to the extent to which information and operations technologies are coming together to share components, functions and staff. Traditionally, the systems, processes and people within IT and OT have been managed, controlled and governed independently of one another. IT/OT convergence promotes a single view of enterprise information and process management to help ensure that every person, sensor, switch or other device has the right information, in the right format, at the right time. It is the integration of previously separate technologies to interact with each other synergistically (Henderson and Venkatraman, 1994). Agarwal and Brem (2015a) reported that IT alone has not provided firms with a sustainable advantage, but that firms have benefited from the integration of strategic planning, organizational learning, and innovative applications of IT. In this context, Gartner (2014) has studied the collaboration of IT and OT and claims that the combination and convergence of these two technologies offers a unique opportunity to companies, with the potential to transform the organization. Achieving IT/OT alignment at the technical



and process level is the first step toward an effective digital business strategy (Gartner, 2014). This has accelerated the development in which operational systems are becoming direct extensions of information systems, and vice versa. In most organizations, there is still too much of a barrier between the IT and the OT functional areas (Bloem et al., 2014a). Getting IT and OT to work together to maximize business efficiency, while avoiding negative consequences, risks and pitfalls in the process makes the task even more challenging. Here we try to assess the alignment of business process redesign and integration that may enable such convergence to happen. The convergence of IT and OT within manufacturing organizations is such an important part of IoT and its adoption that we use it as the basis of our model. The scales adopted for IT/OT convergence are the degree of IT flexibility in the organization (Olawuyi and Friday, 2012), the extent of IT integration (Blackstock and Lea, 2014a), inter-organizational coordination (Derhamy et al., 2017; Derhamy et al., 2015a), and the rate of obsolescence of operational processes (Kagermann et al., 2013). To that effect, we propose the following hypothesis:

**H5.** IT/OT convergence is positively related to the adoption of IoT in manufacturing.

### 3.6. IoT adoption

Although IoT adoption in manufacturing is positioned to have a significant impact on manufacturing activities, most manufacturing companies have limited understanding of IoT or how to apply it to their business (Brandt, 2015). The adoption and deployment of IoT technologies is leading to architectural changes to industrial automation and control systems which includes greater connectivity to industrial systems (Boyes et al., 2018; Kamble et al., 2018). IoT adoption will depend on the likelihood that top management links IoT investment with gaining competitive advantage (Kuan and Chau, 2001), having a strong working knowledge of the adoption and deployment of IoT technologies (Oliveira and Martins, 2011; Brown and Grant, 2005), ensuring that IoT adoption is consistent with company's security and privacy policies [reference], and having a culture that promotes IT adoption in general (Chau and Tam, 1997; Blackstock and Lea, 2014b; Derhamy et al., 2015b). Our preliminary tests showed that these variables show good reliability and validity as indicators for IoT adoption. Getting processes and the people involved to work together to maximize business efficiency in an integrated way is the ultimate goal of an effective IoT adoption. To this effect, we define IoT enablers as IT/OT governance, IT infrastructure, interoperability and staff collaboration, and we offer the offer the following hypothesis:

**H6.** IT/OT convergence mediates the effect between IoT enablers and IoT adoption in manufacturing.

## 4. Methodology

In order to test our hypotheses, we collected the opinions and attitudes of a large set of manufacturing executives by developing an instrument consisting of a set of indicator variables for the exogenous variables in our model shown in Fig. 2. The indicants are based on discussions we found in the literature for each factor. The final survey was administered by a commercial vendor that specializes in assessments, survey development, and analysis. The company administered the survey to a pool of senior manufacturing executives involved in IoT adoption decisions and usage. Twenty-one percent of the respondents were CEO/chairman of their companies, 34% were CIO or IT Directors, and 35% were senior IT specialists. The estimated tenure of the respondents with the companies average 9.6 years with more than 30% having more than 10 years of experience with the companies. Sixty-one per-

**Table 1**  
Sample demographics

| Which best describes your company's knowledge of IoT             | Number | Percent |
|------------------------------------------------------------------|--------|---------|
| Fully conversant on all aspects of IoT                           | 50     | 21%     |
| Understand most of the concepts of IoT                           | 96     | 40%     |
| Understand the basic principles of IoT                           | 70     | 29%     |
| Little knowledge of IoT                                          | 3      | 3%      |
| No knowledge of IoT                                              | 15     | 6%      |
| Total                                                            | 239    | 100%    |
| Which of the following business sectors is your company?         | Number | Percent |
| Shipping/logistics/transportation                                | 25     | 7%      |
| Warehousing                                                      | 24     | 7%      |
| Document management                                              | 11     | 3%      |
| Packaging                                                        | 13     | 4%      |
| Manufacturing (Additive)                                         | 42     | 12%     |
| Manufacturing (Industrial)                                       | 193    | 57%     |
| Fabrication/stamping                                             | 13     | 4%      |
| Food processing                                                  | 3      | 1%      |
| Other                                                            | 15     | 4%      |
| Total                                                            | 339    |         |
| Challenges in adopting IoT in your company                       | Number | Percent |
| Identifying opportunities/benefits of IoT                        | 97     | 16%     |
| Network capabilities to handle IoT                               | 95     | 15%     |
| Budget/resources to develop and expand IoT                       | 98     | 16%     |
| Incorporating smart devices or embedded intelligence in products | 63     | 10%     |
| Adapting existing technologies                                   | 110    | 18%     |
| Clear understanding of customer needs/value                      | 71     | 11%     |
| Knowledge of where/how to get started with IoT enabled products  | 55     | 9%      |
| Interoperability of inter-connected devices                      | 35     | 6%      |
| Total                                                            | 624    |         |

Note: Some of the respondents selected multiple choices.

cent of the respondents have knowledge that ranged from being fully conversant to highly knowledgeable with IoT systems with only 18% of the respondents that have little to no knowledge of IoT. In terms of industry representation, 57% of the respondents are from subtractive manufacturing companies, whereas only 12% were from additive manufacturing companies. The biggest challenge facing IoT adopters from the respondents' viewpoints was adapting existing technologies to new technologies, followed by acquiring resources to invest in new technologies and establishing a business case for the IoT adoption. A summary of the demographic information is presented in Table 1.

## 5. Control Variable

In this study, we assume that large firms have a better opportunity to adopt IoT than smaller firms due to their ability to garner resources and to improve efficiency due to economies of scale (Hitt et al., 2002). We therefore included firm size as a control variable, and measured it using the total number of full-time equivalent employees. Including firm size in the model controls for factors such as relative bargaining power and size of the resource base that can affect technology adoption (Zaheer and Venkatraman, 1994). Larger manufacturers may be more willing to make investments in advanced technology such as IoT. They may also have more experience utilizing information technology within their organization and throughout their supply chain. Benefits to larger companies may therefore be systematically higher than those to smaller companies (Lee and Lee, 2015). However, larger companies may have existing IT and manufacturing operations that need considerable

redesign to integrate with newer and more advanced systems such as IoT. Including firm size as a control variable allows us to isolate these extraneous effects that might otherwise confound the results of the study.

## 6. Analysis and Results

### 6.1. The Measurement Model

PLS is a statistical modelling technique that is well suited in exploratory research where the emphasis is placed on attempting to develop new theories. Unlike its confirmatory counterpart, covariance-based structural equation modelling, it is a variance-based regression procedure that utilizes ordinary least squares (OLS) to explain the variance in the dependent variables of the model (Rai et al., 2006). When the emphasis is more on exploration than confirmation, PLS is an attractive alternative to covariance-based SEM (Hair et al., 2014). Our model is formulated such that variance in the constructs are reflected by changes in our indicator variables. To test our model for reliability, we used coefficient alpha, because it provides an adequate test for internal consistency, which is seen as a good substitute for reliability. This is especially true for instruments that gather results from large groups of disparately located participants (Hair et al., 2014; Pedhazur et al., 1991). Values for coefficient alpha in our model range from 0.701 to 0.913. These are greater than the generally accepted threshold value of 0.70 (Bernstein and Nunnally, 1994). We also checked composite reliability, which takes into account indicators that have different loadings. As shown in Table 4, both coefficient alpha and the composite reliability exceeded the threshold values of 0.70, which provides strong support for the reliability of the constructs.

To assess validity, we computed both convergent and discriminant validity. Convergent validity signifies that a set of indicators represents one and the same underlying construct. We computed the average variance extracted (AVE) to check for convergent validity. A value of 0.50 or greater indicates sufficient convergent validity, implying that the construct is able to explain more than 50% of the variance (Götz et al., 2010). As shown in Table 2, AVE values range from 0.446 to 0.630. This provides support for convergent validity of all the constructs except staff collaboration, but because staff collaboration was just slightly below the minimum, we suspected that there might be an interaction effect (Table 3).

Discriminant validity postulates that a construct shares more variance with its assigned indicators than with any other construct (Fornell and Larker, 1981). For this to hold, the square root of the AVE of a given construct should be greater than the correlation between the construct and all other constructs in the model. The square of the diagonal in Table 4 range from 0.668 to 0.793 which are all greater than the inter-correlations between the constructs.

Given the results of these tests, the measurement model is considered adequate. Having determined that the measurement model meets the evaluation criteria, we now evaluate the structural model to determine its accuracy and predictive capability.

### 6.2. The Structural Model

The evaluation of the structural model tests the hypotheses. The structural model is evaluated using two metrics: the Pearson's coefficient of determination ( $R^2$ ), and the significance of the path coefficients.  $R^2$  measures the proportion of the variance in the dependent variable that is explained by the independent variables. Values of  $R^2$  greater than 0.25 show substantial effects of the model (Oesterreich and Teuteberg, 2015). The  $R^2$  values for the IT/OT convergence and the IoT adoption are 0.429 and 0.355 respectively. This implies that a significant portion of the variances

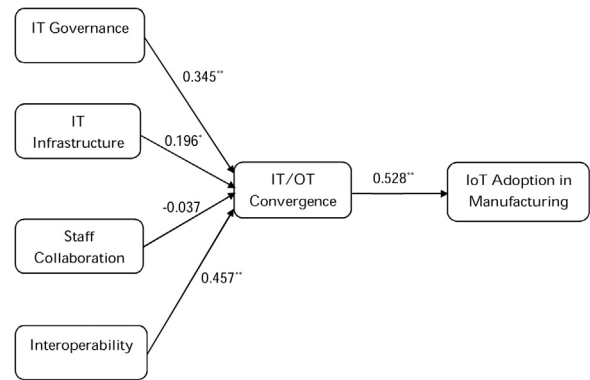


Fig. 3. Model Results (\*\* $p < 0.001$ ; \* $p < 0.05$ ).

of the endogenous variables are explained by the models. Having established that the models have predictive power as shown by the  $R^2$ , we now use the structural equation modelling (PLS) to test the hypothesized relationships. PLS uses available data to estimate the path relationships in the model with the objective of minimizing the error terms of the endogenous constructs (Hair et al., 2014). The predictive capabilities of the adjusted models were also evaluated by examining the change in  $R^2$  using Cohen's indicator or the effect size ( $f^2$ ) and Cross-validated Redundancy or Stone-Geisser indicator ( $Q^2$ ). According to Hair et al. (2014),  $f^2$  and  $Q^2$  values for dependent variables should be greater than zero. Further, they suggest that values of 0.02, 0.15, and 0.35 demonstrate small, medium and large amounts of predictive relevance. The  $f^2$  effect size measures the impact of each predictor variable on the dependent variable. The results indicate that  $f^2$  values range from 0.127 to 0.273 implying that the exogenous variables have medium to large effect and are useful in predicting the endogenous variables.  $Q^2$  also measures the predictive relevance of the model where values below 0 indicate lack of predictive relevance. The results also show  $Q^2$  values of 0.302 for the IT/OT convergence and 0.265 for IoT adoption. Clearly the models have adequate predictive capabilities.

We modeled the second order construct, IoT Adoption, with a separate set of indicator variables taken from the literature on technology adoption as mentioned previously. To build a model in PLS, we compute a set of weights that reflects the covariance structure between the predictor and the response variables. To estimate the significance of path coefficients, we used a bootstrapping technique by re-estimating the path coefficients of 2,000 sub-samples. The resulting matrix of parameter estimates was used to compute parameter means, standard errors, significance of path coefficients, indicator loadings, and indicator weights. This approach is consistent with recommended practices for estimating significance of path coefficients and indicator loadings (Lohmöller, 1984).

The results of our analysis for the structural model are presented in Fig. 3. We found that both IT governance and Interoperability significantly affect IT/OT Convergence with a value of 0.345 and 0.457 respectively ( $p < 0.01$ ). Additionally, we found that IT infrastructure also significantly affected IT/OT Convergence with a value of 0.196 ( $p \leq 0.05$ ). We therefore sustain hypotheses  $H_1$ ,  $H_2$ , and  $H_4$ . However, we found a non-significant relationship between staff collaboration and IT/OT convergence with a weak path coefficient of -0.029 ( $H_3$ ), and therefore we reject hypothesis  $H_3$ . However, because the Staff Collaboration construct had a marginal average variance extracted in our initial tests for validity, we suspected that an interaction effect might be the reason for this non-significant finding. We therefore tested to see if any indirect effects were present. We found that there is a significant interaction between staff collaboration and IT governance as shown in Table 5. Furthermore, we found a statistically significant relationship between

**Table 2**  
Measurement of Constructs.

| Latent Construct    | Indicator Variables                                                                                                                                                                                                                                                                                                                                                | Cronbach alpha | Composite Reliability | Average Variance Extracted (AVE) |
|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|-----------------------|----------------------------------|
| IT Governance       | Strategic redefinition of organizational units Top management role in enabling IT//OT convergence Pattern of managerial responsibilities and control in the organization Business case for IT investments                                                                                                                                                          | 0.833          | 0.834                 | 0.626                            |
| IT Infrastructure   | Decentralizing systems across the organization Adapting existing technologies to new technologies Rapidly changing technology Cooping with frequent changes in production technology                                                                                                                                                                               | 0.752          | 0.769                 | 0.531                            |
| Staff Collaboration | Resolving technical operations issues Recruiting appropriate talent needs of the enterprise Upgrading legacy operations systems Talents/skills to links operations data with advanced analytics                                                                                                                                                                    | 0.701          | 0.706                 | 0.446                            |
| Interoperability    | Technical integrations across disparate systems Compatible IT systems with partners Seamlessly connected IT systems with partners Leveraging IT capabilities among systems across the supply chain IT systems similar to supply chain partners                                                                                                                     | 0.872          | 0.872                 | 0.630                            |
| IT/OT Convergence   | IT flexibility IT integration Inter-organization coordination Rate of obsolescence of operational processes                                                                                                                                                                                                                                                        | 0.813          | 0.819                 | 0.532                            |
| IoT Adoption        | Top management links IoT investment to gaining competitive advantage Strong working knowledge of IoT exists in my company IoT adoption enables quick data capture and analysis IoT adoption is consistent with company's beliefs and values IoT adoption is consistent with company's security policies IoT adoption is consistent with company's privacy policies | 0.913          | 0.908                 | 0.526                            |

**Table 3**  
Outer factor loading for the constructs.

| Item/Construct | Governance | Interoperability | Infrastructure | Collaboration | IT/OT Convergence | IoT Adoption |
|----------------|------------|------------------|----------------|---------------|-------------------|--------------|
| GO1            | 0.795      |                  |                |               |                   |              |
| GO2            | 0.850      |                  |                |               |                   |              |
| GO3            | 0.727      |                  |                |               |                   |              |
| GO4            | 0.705      |                  |                |               |                   |              |
| OP1            |            | 0.810            |                |               |                   |              |
| OP2            |            | 0.831            |                |               |                   |              |
| OP3            |            | 0.845            |                |               |                   |              |
| OP4            |            | 0.875            |                |               |                   |              |
| OP5            |            | 0.686            |                |               |                   |              |
| IF1            |            |                  | 0.661          |               |                   |              |
| IF2            |            |                  | 0.883          |               |                   |              |
| IF3            |            |                  | 0.902          |               |                   |              |
| ST1            |            |                  |                | 0.837         |                   |              |
| ST2            |            |                  |                | 0.874         |                   |              |
| ST3            |            |                  |                | 0.887         |                   |              |
| CV1            |            |                  |                |               | 0.780             |              |
| CV2            |            |                  |                |               | 0.846             |              |
| CV3            |            |                  |                |               | 0.888             |              |
| CV4            |            |                  |                |               | 0.684             |              |
| AD1            |            |                  |                |               |                   | 0.801        |
| AD2            |            |                  |                |               |                   | 0.792        |
| AD3            |            |                  |                |               |                   | 0.802        |
| AD4            |            |                  |                |               |                   | 0.771        |
| AD5            |            |                  |                |               |                   | 0.780        |
| AD6            |            |                  |                |               |                   | 0.786        |

**Table 4**  
Descriptive statistics and correlation matrix.

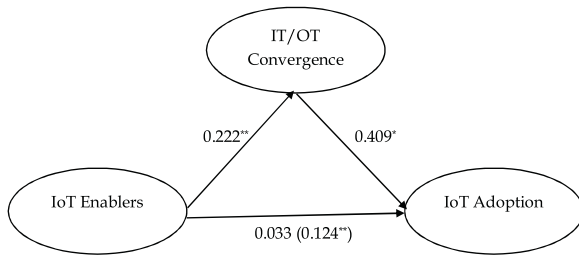
|    |                     | Mean  | Std Dev | 1            | 2            | 3            | 4            | 5            | 6            |
|----|---------------------|-------|---------|--------------|--------------|--------------|--------------|--------------|--------------|
| 1. | IT Governance       | 4.954 | 1.067   | <b>0.791</b> |              |              |              |              |              |
| 2. | Interoperability    | 5.034 | 0.936   | 0.462        | <b>0.793</b> |              |              |              |              |
| 3. | Staff Collaboration | 5.246 | 0.943   | 0.306        | 0.445        | <b>0.668</b> |              |              |              |
| 4. | IT Infrastructure   | 5.122 | 0.95    | 0.438        | 0.472        | 0.279        | <b>0.729</b> |              |              |
| 5. | IT/OT Convergence   | 5.078 | 1.047   | 0.512        | 0.533        | 0.284        | 0.479        | <b>0.729</b> |              |
| 6. | IoT Adoption        | 5.156 | 0.914   | 0.183        | 0.315        | 0.242        | 0.184        | 0.245        | <b>0.725</b> |

\*\*Correlation is significant at the 0.01 level (2-tailed). Diagonal in bold is the square root of AVE.

**Table 5**  
OLS Regression results of the IoT enablers on IT/OT Conversion.

| Construct                           | $\beta$ -value <sup>a</sup> | Std. Error | t-value | p-value |
|-------------------------------------|-----------------------------|------------|---------|---------|
| Intercept                           | −0.79                       | 0.873      | −0.905  | 0.366   |
| IT Governance                       | 0.735                       | 0.019      | 2.459   | 0.000   |
| IT Infrastructure                   | 0.503                       | 0.205      | 5.606   | 0.015   |
| Staff Collaboration                 | −0.260                      | 0.163      | 4.215   | 0.112   |
| Interoperability                    | 0.415                       | 0.074      | −1.595  | 0.000   |
| IT Governance x Staff Collaboration | 0.090                       | 0.039      | −2.332  | 0.021   |

<sup>a</sup> Unstandardized parameter estimate;  $R^2 = 0.075$ ;  $F$ -value = 116.14.



**Fig. 4.** Mediation effect of the IT/OT Convergence.

IT/OT convergence and IoT adoption with a value of 0.528 ( $p \leq 0.01$ ) thus confirming hypothesis  $H_5$ . Our results confirm the predictive role of the IoT enablers (IT governance, IT infrastructure, and interoperability) on IT/OT convergence and that this in turn has a positive influence on IoT adoption.

### 6.3. Mediation Effect of IT/OT Convergence

Although we found a significant positive and direct relationship between IT/OT convergence and IoT adoption, we decided to run a mediation analysis to test the extent to which IT/OT convergence might act as a mediator of the IoT Enablers on IoT adoption. To do this, we compared our original model against one that allows a direct effect from the IoT Enablers to IoT adoption and an indirect effect from the IoT enablers to IoT adoption. Such a model would capture any mediating effects of IT/OT convergence. This model is depicted in Fig. 4. These tests are considered complementary tests of mediation (Subramanian, 2004) because the nested model assesses the additional explanatory power of competing models, while the mediation analysis provides information on the significance of mediation effects. We ran the partial and full mediation analysis using Process, v3.0 technique in SPSS (Preacher and Hayes, 2004). We standardized the scores for the IoT enablers' constructs and then used summative measures to develop a single construct for the IoT enablers based on the three significant constructs (IT governance, IT infrastructure, and interoperability). We assess the significance of the mediation effect of IT/OT convergence on the relationship between IoT enablers and IoT adoption. The analysis is based on the path coefficients and the standard errors of the direct paths between the IoT enablers (exogenous variables), IT/OT convergence (mediation effect), and IoT adoption (endogenous variable). The magnitude of mediation is assessed by the unstandardized path coefficients between IoT enablers and IT/OT convergence, and between IT/OT convergence and IOT adoption and the resulting  $R^2$ . The magnitude of mediation effect between IoT enablers and IT/OT convergence as a mediator was significant with a path coefficient of 0.222 and an  $R^2$  of 0.344. The path between IT/OT convergence and IoT adoption was stronger with a significant ( $p < .01$ ) path coefficient of 0.409 and an  $R^2$  of 0.232. The path from IoT enablers and IoT adoption was not significant ( $\beta = 0.033$ ) resulting in a partially mediated effect. However, the path coefficient without the mediation between IoT enablers and IoT adoption was significant ( $\beta = 0.124$ ). The indirect effect of IT/OT convergence

on IoT adoption is significant at 0.091. The results, shown in Fig. 4 imply that IT/OT convergence is a strong mediator of IoT enablers and IoT adoption, stronger than the effect of the IoT enablers alone on IoT adoption.

### 7. Analysis Triangulation

Considering the exploratory nature of our study, we conducted a triangulation to further validate our results. An analysis triangulation is the use of multiple methods of analyzing the same set of data for validation purposes (Kimchi et al., 1991). We subjected our dataset to an OLS regression analysis, testing the effect of the IoT enablers (IT governance, interoperability, infrastructure, and staff collaboration) on IT/OT convergence and later, tested how IT/OT convergence affects IoT adoption. The results corroborated the findings obtained using PLS analysis and confirm strong and significant relationships except for staff collaboration. We also tested for the interaction effect between staff collaboration and each of the IoT enablers. The results show non-significant relationship in the interaction effect except for IT governance and staff collaboration. The results are shown in Table 5. This implies that staff collaboration by itself may not significantly affect IT/OT convergence but its interaction with IT governance does play an important role. This result is not surprising as it is in agreement with a body of research highlighting the important role collaboration plays in governance systems (Wondolleck and Yaffee, 2000; Born and Genskow, 2001; Leach et al., 2002) The implication of this finding is that companies wanting to adopt IoT successfully should carefully consider the impact that the collaboration of their IT and OT staffs will have on IT governance, because such policies will affect the convergence of their IT and OT systems.

### 8. Discussion

The results suggest that IoT enablers in the form of IT infrastructure, IT governance, interoperability, and to a lesser extent, staff collaboration, positively impact IT/OT convergence, which in turn positively influences IoT adopt in manufacturing organizations. IT/OT convergence is the integration of information technology (IT) systems used for data-centric computing with operational technology (OT) systems used to monitor events, processes and devices and make adjustments in enterprise and industrial operations. This is the amalgamation of two independent, historical and economic developments – that of advanced information technology, with predictive data analytics, smartphones and traditional administrative automation, among other things – and that of the impressive machinery and automation that have been designed and developed since the start of the industrial revolution (Bloem et al., 2014b). Advanced data analytical tools such as visualization, scenario analysis, and predictive learning algorithms, are typically referred to as information technology (IT) whereas advanced physical technologies that include robotics, drones, additive manufacturing (3D printing), and autonomous vehicles, are typically called operational technology (OT). Our study has shown evidence that IT/OT convergence is fundamental to the adoption of IoT sys-



tems, because it helps to integrate the operational technology that creates massive amounts of data and information systems, which forms the infrastructure needed to process the data. This convergence will produce new standards and represents a paradigm shift in the implementation of new IoT systems. This new paradigm is characterized by the integration of automation, communications and networking in industrial environments between IT and OT (Gartner, 2014). Traditionally, the two technologies – operational technology (OT) and information technology (IT) – have operated independently from one another; however, technological advancements that businesses are experiencing have increased the overlap and convergence of these two areas. Manufacturing organizations are investing heavily in the integration and alignment of these technologies and expect to benefit in several ways from the convergence. Benefits that come from managing IT/OT convergence include optimized business processes, enhanced information for better decisions, improved maintenance/uptime, reduced manufacturing costs, lower risks and shortened project timelines (Brandt, 2015; Gartner, 2014). IT/OT convergence allow companies to seamlessly and automatically integrate data from both internal and external sources in order to better manage their global supply chain. This paper presents the results of an empirical study that identifies four IoT enablers from the literature that leads to IT/OT convergence. The study further shows that with technology convergence, this leads to an effective IoT adoption. To corroborate the findings, Bi et al. (2014) discovered that the emerging IoT infrastructure supports information systems of next-generation manufacturing enterprises effectively.

IoT has become the foundation for connecting devices, sensors, actuators, and other smart technologies (Uckelmann et al., 2011). These intelligent systems can enable rapid manufacturing of new products, dynamic response to product demand, and real-time optimization of supply chain networks. Yet implementing IoT presents challenges, which include ensuring that disparate systems are interoperable, attaching smart objects to interconnected networks and ensuring that these systems are built with appropriate governance structure in place that includes trust, security, privacy of users and organizations in mind (Bi et al., 2014). Therefore, by investigating the aforementioned drivers of IoT adoption, we made contributions to the technology-enabled manufacturing business practices. Our findings provide significant managerial implications by revealing that the convergence of IT/OT systems is enabled by four factors and that this can play a major role in the effective adoption of IoT systems in manufacturing. This convergence in turn improves systems connectivity, which is necessary to allow companies to effectively use of their IoT systems. Effective IT/OT convergence which allows organizations to collect, integrate, analyse and disseminate information greatly improves the chances of IoT adoption in manufacturing organizations.

## 9. Conclusions and Limitations

This study investigates the role of IT/OT convergence on the adoption of IoT in manufacturing operations using data collected from 239 manufacturing firms in the United States and makes significant contributions to both theory and practice. This is the first study to empirically study the adoption of IoT systems in manufacturing. The major contributions are discussed next.

Industrial organizations are investing heavily in the integration and alignment of these technologies and expect enormous benefits from the IT and OT convergence (Agarwal and Brem, 2015b). The convergence of these two technologies offer a unique opportunity to companies with the potential to revolutionize the manufacturing sector (Gartner, 2014). Through an extensive literature review, we identified four major factors as key enablers to IT/OT conver-

gence. We tested the proposed framework using the manufacturing dataset and found that IT infrastructure, IT governance, and systems interoperability all have direct and positive relationship with IT/OT convergence. Contrary to past research, we found no empirical evidence to support a direct effect between staff collaboration and IT/OT convergence. Staff collaboration while considered as an important factor, did not play directly into IT/OT convergence. We investigated this further and conducted an interactions analysis among the IoT enablers and found that staff collaboration becomes significant when interacted with IT governance. The implication is that staff collaboration when developed within the context of the IT governance has significant impact on IT/OT convergence. Once established, we then tested the relationship between IT/OT convergence and IoT adoption and found a stronger and significant relationship between the convergence of IT/OT and IoT adoption. This is a significant contribution in the sense that IT/OT convergence serves as a strong enabler to IoT adoption. To confirm this finding, we tested for the mediating effect of IT/OT convergence on IoT adoption and found that a partial, but significant mediation effect exists. These results become critical as more companies invest in IoT systems to try and improve their competitive position in the marketplace. IT/OT convergence is expected to enable companies to offer value added services like advanced analytics and real time intelligence to their customers. The findings enrich our understanding of the enabling factors to IoT adoption.

Although this study makes both theoretical and practical contributions to the literature, it has a number of limitations. Given the anecdotal evidence on the benefits of IoT systems, we did not test the effect of the IoT adoption on firm performance. We limited our investigation to empirically establish factors that promote successful IoT adoption. Now that the IoT enablers have been identified and tested, future studies can test for the performance of the organization given the IoT adoption. Second, the IoT enablers were those identified from current literature and may not be exhaustive. Further studies should explore other enabling factors, especially as IoT systems continue to evolve, that would lend themselves to effective IT/OT convergence which will result in effective IoT adoption. Third, the survey frame is limited to US manufacturing companies. Future studies may investigate IoT adoption on a global scale to test and validate the generalizability of the model in other countries. Cross-cultural investigation of IoT adoption may reveal other pertinent factors. The relationships between the IoT enablers, the IT/OT convergence, and the IoT adoption are assumed to be a linear, second-order model. Future studies could explore alternative models to better establish the relationships among these IoT adoption factors. Finally, we collected cross-sectional data from a single industry (manufacturing). Future studies could investigate the role IT/OT convergence on IoT adoption using longitudinal data from other industries to help test the generality of our findings.

## 10. Managerial implications

Studies have shown that IT investment alone has not provided firms with sustainable competitive advantage but that firms have benefited from strategic planning, IT integration, organization learning and innovative applications of IT (Tippens and Sohi, 2003; Teece et al., 1997; Bharadwaj, 2000). Unfortunately, many companies fail to grasp the emphasis on “how” information technology is used to facilitate transformative improvements and have instead focused on the technology itself (Fawcett et al., 2011). Consequently, these companies have failed to use IT to leverage complementary competencies to gain competitive advantage (Frohlich, 2002). The convergence of IT and OT and the enabling factors that facilitate this convergence are rooted in both technological and strategic imperatives that make the IoT adoption

a more holistic experience. In most organizations, there is still a significant barrier between the two technologies, thus making convergence much more difficult. However, IoT, brings with it an interdependence and synergy between the two technologies that can accelerate innovation. As this occurs, the underlying technologies of OT systems, such as platforms, software, security, and communications, are becoming more like IT systems, and IT is supporting OT in building standards, enterprise architecture (EA), support and security models, software configuration practices, and information and process. The future of this convergence, as demonstrated in the effective adoption and installation of IoT systems, looks bright.

### Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

### References

- Ben-Daya, M., Hassini, E., Bahroun, Z., 2017. Internet of things and supply chain management: a literature review. *International Journal of Production Research*, <http://dx.doi.org/10.1080/00207543.2017.1402140>.
- Ellis, S., Morris, H., Santagata, J., White Paper 2015. IoT-enabled analytic applications revolutionize supply chain planning and execution. International Data Corporation (IDC) [www.idc.com](http://www.idc.com).
- Brandt, J., 2015. The internet of things has finally arrived: Unfortunately, manufacturers aren't ready. The MPI Group, Executive Summary Report., pp. 1–9.
- Hofmann, E., Rüscher, M., 2017. Industry 4.0 and the current status as well as future prospects on logistics. *Computers in Industry* 89, 23–34.
- Gubbi, J., Buyya, R., Marusic, S., Palaniswami, M., 2013. Internet of things (IoT): A vision, architectural elements, and future directions. *Future Generation Computer Systems* 29 (7), 1645–1660.
- Verdouw, C., Beulens, A., van der Vorst, J., 2013a. Virtualization of floricultural supply chains: A review from an Internet of Things perspective. *Computers and Electronics in Agriculture* 99, 160–175.
- Borgia, E., 2014. The internet of things vision: Key features, applications and open issues. *Computer Communications* 54, 1–31.
- Xu, L., He, W., Li, S., 2014. Internet of things in industries: A survey. *IEEE Transactions on Industrial Informatics* 10 (4), 2233–2243.
- Gurria, A., 2017. Remarks on The Impact of scientific and technological advances and innovation on the transition of traditional industries and the development of new industries in China, 10–12 September, Beijing, China.
- Bi, Z., Xu, L., Wang, L.C., 2014. Internet of things for enterprise systems of modern manufacturing. *IEEE Transactions on Industrial Informatics* 10 (2), 1537–1546.
- Lee, I., Lee, K., 2015. The Internet of Things (IoT): Applications, investments, and challenges for enterprises. *Business Horizon* 58 (4), 431–440.
- Rodrigue, M., Libbey, R., Mondal, S., Carbeck, J., Michalik, J., 2017. Exponential technologies in manufacturing through technology, talent, and innovation ecosystem, Council of Competitiveness. Deloitte and Singularity University Publication.
- Gold, J., Retrieved from: 2017. Skunk works with tractors: Inside John Deere's IoT-innovation unit. *Network World* <https://www.networkworld.com/article/3198744/internet-of-things/a-skunk-works-with-tractors-inside-john-deere-s-high-tech-iot-innovation-unit.html>.
- Horwitz, L., Retrieved from: 2018. UPS IoT strategy is fueled by analytics to bring business insight. <https://www.cisco.com/c/en/us/solutions/internet-of-things/iot-strategy.html>.
- Gartner, I.T., February 2014 2014. Operational Technology: Convergence, Alignment and Integration. <https://www.gartner.com/newsroom/id/2905717>.
- Porter, M., Heppelmann, J., October 2015. How smart, connected products are transforming companies, *Harvard Business Review*., pp. 3–17.
- Tornatzky, L., Fleischer, M., Chakrabarti, A., 1990. *Processes of Technological Innovation*. Lexington Books.
- Rogers, E., 2003. *Diffusion of Innovations*, 5th ed. The Free Press, New York, NY.
- Iacovou, C., Benbasat, I., Dexter, A., 1995. Electronic data interchange and small organizations: Adoption and impact of technology. *MIS quarterly*, 465–485.
- World Economic Forum (WEF) Deloitte Touche Tohmatsu (Firm), 2012. *The Future of Manufacturing: Opportunities to Drive Economic Growth*. World Economic Forum, Geneva, Switzerland.
- Buyukozkan, G., Gocer, F., 2018. (Digital supply chain: Literature review and proposed framework for future research. *Computers in Industry* 97, 157–177.
- Schlick, J., Ferber, S., Hupp, J., 2013. *IoT Applications – Value Creation for Industry*. River Publisher, Aalborg.
- Verdouw, C., Vucic, N., Sundmaeker, H., Beulens, A., 2013b. Future internet as a driver for virtualization, connectivity and intelligence of agri-food supply chain networks. *International Journal on Food System Dynamics* 4 (4), 261–272.
- Ng, I., Scharf, K., Pogrebnia, G., Maull, R., 2015. Contextual Variety, Internet-of-Things and the Choice of Tailoring over Platform: Mass Customisation Strategy in Supply Chain Management. *International Journal of Production Economics* 159, 76–87.
- Yu, J., Subramanian, N., Ning, K., Edwards, D., 2015. Product Delivery Service Provider Selection and Customer Satisfaction in the Era of Internet of Things: A Chinese E-Retailers' Perspective. *International Journal of Production Economics* 159, 104–116.
- Decker, C., Berchtold, M., Chaves, L.W.F., Beigl, M., Roehr, D., Riedel, T., Beuster, M., Herzog, T., Herzog, D., 2008. Cost-Benefit Model for Smart Items in the Supply Chain. *The Internet of Things Lecture Notes in Computer Science*, Vol. 4952., pp. 155–172.
- Liu, Y., Han, W., Zhang, Y., Li, L., Wang, J., Zheng, L., 2016. An Internet-of-Things Solution for Food Safety and Quality Control: A Pilot Project in China. *Journal of Industrial Information Integration* 3 (3), 1–7 (2016).
- Tan, L., Wang, N., 2010. Future internet: The internet of things. *IEEE In: 2010 3rd international conference on advanced computer theory and engineering (ICACTE)*, Vol. 5, pp. V5–376.
- Haller, S., Karnouskos, S., Schroth, C., 2009. The Internet of Things in an Enterprise Context. *Lecture Notes in Computer Science Future Internet – FIS*, Vol. 5468., pp. 14–28 (2009).
- Bandyopadhyay, D., Sen, J., 2011. Internet of things: Applications and challenges in technology and standardization. *Wireless Personal Communications* 58, 49–59.
- Liu, P., Huang, S., Mokasdar, A., Zhou, H., Hou, L., 2014. The impact of additive manufacturing in the aircraft spare parts supply chain: supply chain operations reference (scor) model-based analysis. *Production Planning & Control* 25 (13–14), 1169–1181.
- Yao, L., Guozhu, Y., Cheng, Y., Hu, Y., 2017. Additive manufacturing technology in spare parts supply chain: a comparative study. *International Journal of Production Research* 55 (5), 1498–1515.
- Ha, S., Stael, K., 2009. Consumer e-shopping acceptance: Antecedents in a technology acceptance model. *Journal of business research* 62 (5), 565–571.
- Hsu, P., Kraemer, K., Dunkle, D., 2006. Determinants of e-business use in us firms. *International Journal of Electronic Commerce* 10 (4), 9–45.
- Kuan, K., Chau, P., 2001. A perception-based model for EDI adoption in small businesses using a technology-organization-environment framework. *Information & Management* 38 (8), 507–521.
- Starbuck, W., 1976. *Organizations and their environments*. Rand McNally, Chicago.
- Oliveira, T., Martins, M., 2011. Literature Review of Information Technology Adoption Models at Firm Level. *The Electronic Journal Information Systems Evaluation* 14 (1), 110–121.
- Li, L., 2006. The effect of information technology implementation on supply chain collaboration. *International Journal of Internet and Enterprise Management* 4 (2), 118–134.
- Chau, P., Tam, K., 1997. Factors affecting the adoption of open systems: An exploratory study. *MIS Quarterly* 21 (1), 1–24.
- Haider, A., 2012. Information and operational technologies governance framework for engineering asset management. In: Mathew, J., Ma, L., Tan, A., Weijnen, M., Lee, J. (Eds.), *Engineering Asset Management and Infrastructure Sustainability*. Springer, London.
- Ye, F., Wang, Z., 2013. Effects of information technology alignment and information sharing on supply chain operational performance. *Computers & Industrial Engineering* 65, 370–377.
- DePietro, R., Wiarda, E., Fleischer, M., 1990. The context for change: Organization, technology and environment. In: Tornatzky, L.G., Fleischer, M. (Eds.), *The Processes of Technological Innovation*. Lexington Books, Lexington, MA, pp. 151–175.
- Loh, L., Venkatraman, N., 1992. Determinants of information technology outsourcing: a cross-sectional analysis. *Journal of management information systems* 9 (1), 7–24.
- De Haes, S., Van Grembergen, W., 2005. IT governance structures, processes and relational mechanisms: Achieving IT/business alignment in a major Belgian financial group. *Proceedings of the 38th Annual Hawaii International Conference on System Sciences*, 237b.
- Khandwalla, P., 1970. *Environment and the organization structure of firms*. McGill University, Montreal, Faculty of Management.
- de Villiers, P., (Unpublished master's thesis) 2015. *An Exploration of the Need for OT Governance and the Adaptation of IT Governance Frameworks to Fulfill this Requirement*. Stellenbosch University, Stellenbosch, South Africa.
- Boynton, A., Jacobs, G., Zmud, R., 1992. Whose responsibility is IT management. *Sloan Management Review* 33 (4), 32–38.
- Duncan, N., 1995. Capturing flexibility of information technology infrastructure: a study of resource characteristics and their measures. *Journal of Information Systems* 12 (2), 37–57.
- Weill, P., 1993. The role and value of information technology infrastructure: some empirical observations. In: Banker, R., Kauffman, R., Mahmood, M.A. (Eds.), *Strategic Information Technology Management: Perspective on Organizational Growth and Competitive Advantage*. Idea Group Publishing, Middleton, PA, pp. 547–572.
- Davenport, T., Linder, J., 1994. Information management infrastructure: the new competitive weapon. *Proceedings of the 27th Annual Hawaii International Conference on Systems Sciences* Vol. 27, 885–899.
- Rockart, J., Earl, M., Ross, J., 1996. Eight imperatives for the new IT organization. *Sloan Management Review* 38 (1), 43–54.
- Broadbent, M., Weill, P., 1997. Management maxim: how business and IT managers can create IT infrastructures. *Sloan Management Review* 38 (3), 77–92.

- Henderson, J., Venkatraman, N., 1994. *Strategic alignment: a model for organizational transformation via information technology*. Oxford University Press, New York, pp. 202–220.
- Agarwal, N., Brem, A., 2015a. Strategic business transformation through technology convergence: implications from General Electric's industrial internet initiative. *International Journal of Technology Management* 67 (2/3/4), 196–214.
- Bloem, J., Doorn, M., Duivestijn, S., Excoffier, D., Maas, R., Ommeren, E., 2014a. The fourth industrial revolution: Things to tighten the link between IT and OT, Sogeti, VINT Report.
- Olawuyi, J., Friday, M., 2012. Technological convergence. *Science Journal of Physics*, 5, <http://dx.doi.org/10.7237/sjp/221>, Article ID sjp-221.
- Blackstock, M., Lea, R., 2014a. IoT interpretability: A hub-based approach. the Proceedings of 2014 Conference in Internet of Things (IOT), 79–84.
- Derhamy, H., Eliasson, J., Delsing, J., 2017. IoT Interoperability: On-demand and low latency transparent multiprotocol translator. *IEEE Internet of Things Journal* 4 (5), 1754–1763.
- Derhamy, H., Eliasson, J., Delsing, J., Priller, P., 2015a. A survey of commercial frameworks for the internet of things. Proceedings of the IEEE 20th Conference of Emerging Technology Factory Automation (ETFA), 1–8.
- Kagermann, H., Wahlster, W., Helbig, J., 2013. Recommendations for implementing the strategic initiative industrie 4.0: Final report of the Industrie 4.0 working group. Acatech-National Academy of Science and Engineering, Munich, Germany.
- Boyce, H., Hallaq, B., Cunningham, J., Warson, T., 2018. The industrial internet of things (IIoT): An analysis framework. *Computers in Industry* 101, 1–12.
- Kamble, S., Gunasekaran, A., Sharma, R., 2018. Analysis of the driving and dependence power of barriers to adopt industry 4.0 in Indian Manufacturing industry. *Computers in Industry* 101, 107–119.
- Brown, A., Grant, G., 2005. (Framing the frameworks: A review of IT governance research. *Communications of the Association for Information Systems* 15, 696–712.
- Blackstock, M., Lea, R., 2014b. IoT interpretability: A hub-based approach. the Proceedings of 2014 Conference in Internet of Things (IOT), 79–84.
- Derhamy, H., Eliasson, J., Delsing, J., Priller, P., 2015b. A survey of commercial frameworks for the internet of things. Proceedings of the IEEE 20th Conference of Emerging Technology Factory Automation (ETFA), 1–8.
- Hitt, L., Wu, D., Xiaoge, Z., 2002. Investment in enterprise resource planning: Business impact and productivity measures. *Journal of Management Information Systems* 19 (1), 71–98.
- Zaheer, A., Venkatraman, N., 1994. Determinants of electronic integration in the insurance industry: an empirical test. *Management science* 40 (5), 549–566.
- Rai, A., Patnayakuni, R., Seth, N., 2006. Firm performance impacts of digitally enabled supply chain integration capabilities. *MIS quarterly* 30 (2), 225–246.
- Hair Jr., F., Sarstedt, M., Hopkins, L., Kuppelwieser, V., 2014. Partial least squares structural equation modeling (PLS-SEM) An emerging tool in business research. *European Business Review* 26 (2), 106–121.
- Pedhazur, E., Pedhazur, J., Schmelkin, L., 1991. *Exploratory factor analysis. Measurement, design and analysis: An Integrated Approach*. Lawrence Erlbaum Associates, Hillsdale NJ, 590, 627.
- Bernstein, I., Nunnally, J., 1994. *Psychometric theory*. McGraw-Hill, New York.
- Götz, O., Liehr-Gobbers, K., Krafft, M., 2010. Evaluation of structural models using the partial least squares (PLS) approach. In: Esposito Vinzi, V., Chin, W.W., Henseler, J., Wang, H. (Eds.), *Handbook of partial least squares: Concepts, methods and applications*. Springer, London, pp. 691–711.
- Fornell, C., Larcker, D., 1981. Structural equation modelling and regression: guidelines for research practice. *Journal of Marketing Research* 18 (1), 39–50.
- Oesterreich, T., Teuteberg, F., 2015. Understanding the implications of digitisation and automation in the context of Industry 4.0: A triangulation approach and elements of a research agenda for the construction industry. *Computers in industry* 83, 121–139.
- Lohmöller, J., 1984. *LVPLS 1.6 program manual: latent variables path analysis with partial least-squares estimation; mit 2 Disketten Version 1.8*. Zentralarchiv für empirische Sozialforschung.
- Subramanian, M., 2004. How do suppliers benefit from information technology use in supply chain relationships? *MIS Quarterly*, 45–73.
- Preacher, K., Hayes, A., 2004. SPSS and SAS procedures for estimating indirect effects in simple mediation models. *Behavior research methods, instruments, & computers* 36 (4), 717–731.
- Kimchi, J., Polivka, B., Stevenson, J., 1991. Triangulation: operational definitions. *Nursing research* 40 (6), 364–366.
- Wondollock, J., Yaffee, S., 2000. *Making collaboration work: Lessons from innovation in natural resource management*. Island Press.
- Born, S., Genskow, K., 2001. *Toward understanding new watershed initiatives—A report from the Madison watershed workshop*. Madison Watershed Workshop, Madison, WI.
- Leach, W., Pelkey, N., Sabatier, P., 2002. Stakeholder partnerships as collaborative policy making: Evaluation criteria applied to watershed management in California and Washington. *Journal of Public Policy Analysis and Management* 21, 645–670.
- Bloem, J., Doorn, M., Duivestijn, S., Excoffier, D., Maas, R., Ommeren, E., 2014b. The fourth industrial revolution: Things to tighten the link between IT and OT, Sogeti, VINT Report.
- Uckelmann, D., Harrison, M., Michahelles, F., 2011. *An architecture approach towards the future Internet of Things*. In: *Architecting the Internet of Things*. Springer-Verlag, Berlin, Germany, pp. 1–24.
- Agarwal, N., Brem, A., 2015b. Strategic business transformation through technology convergence: implications from General Electric's industrial internet initiative. *International Journal of Technology Management* 67 (2/3/4), 196–214.
- Tippens, J., Sohi, S., 2003. IT competence and firm performance: is organizational learning a missing link? *Strategic Management Journal* 24 (8), 745–761.
- Teece, D., Pisano, G., Shuen, A., 1997. Dynamic capabilities and strategic management. *Strategic Management Journal* 18 (7), 509–533.
- Bharadwaj, S., 2000. A resource-based perspective on information technology capability and firm performance: an empirical investigation. *MIS Quarterly* 24 (1), 169–196.
- Fawcett, S., Wallin, C., Allred, C., Fawcett, A., Magnan, G., 2011. Information technology as an enabler of supply chain collaboration: A dynamic-capabilities perspective. *Journal of Supply Chain Management* 47 (1), 38–59.
- Frohlich, M., 2002. E-Integration in the Supply Chain: Barriers and Performance. *Decision Sciences* 33 (4), 537–556.
- Aloi, G., Caliciuri, G., Fortino, Gravina, R., Pace, P., Russo, W., Savaglio, C., 2018. Enabling IoT interoperability through opportunistic smartphone-based mobile gateways. *Journal of Network and Computer Applications* 81, 74–84.
- Luo, Y., Duan, Y., Li, W., Fortino, G., 2018. A Novel Mobile and Hierarchical Data Transmission Architecture for Smart Factories. *IEEE Transactions on Industrial Informatics* 14 (8), 3534–3546.